SFIEET: Spectral Frequency-Induced Edge Enhancement Transformer for Hyperspectral Change Detection

Le Sun, *Senior Member, IEEE*, Bingcheng Shi, Yuhui Zheng, *Member, IEEE*, Liang Xiao, *Senior Member, IEEE*, and Yulin Cao

***Abstract*—Deep learning (DL) methods have achieved remarkable performance in hyperspectral image (HSI) change detection (CD). However, they still face two primary limitations. First, most approaches treat all spectral channels equally, overlooking the varying importance of spectral information. Second, they often struggle to clearly delineate edges in low-resolution images, resulting in suboptimal feature extraction and increased false or missed detections. To address these challenges, we propose a novel spectral frequency-induced edge enhancement Transformer (SFIEET) network to enhance the accuracy and robustness of HSI CD. The proposed SFIEET framework comprises three key modules. The spectral frequency and spatial-domain fusion (SFSDF) module implements an adaptive frequency-domain attention mechanism based on grouped discrete cosine transform (DCT), dynamically focusing on channels rich in weak semantic information. Simultaneously, it integrates multiscale contextual spatial information to capture morphological changes such as texture fragmentation, effectively enhancing recognition of structural changes. The sharpening and Laplace-guided edge enhancement (SLGEE) module integrates multiscale pooling with the Laplacian operator to fuse edge features across spectral bands, enabling precise boundary localization and more reliable detection in blurred regions. Finally, the depthwise convolution-embedded lightweight Transformer (DW-ELT) module incorporates depthwise separable convolutions into the Transformer architecture to enhance local feature modeling, while gated linear units (GLUs) strengthen nonlinear representations, making change-region features more discriminative. Extensive experiments on three benchmark datasets demonstrate that the proposed SFIEET consistently outperforms existing state-of-the-art methods. The source code is available at https://github.com/bcshi83/SFIEET.git for the sake of reproducibility.**

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I. Introduction

HYPERSPECTRAL images (HSIs) have rich spectral and spatial information compared with conventional optical images [1]. With the rapid development of remote sensing technology, quantities of HSI datasets are emerging. This allows different kinds of applications based on HSIs to flourish, such as change detection (CD) [2], [3], anomaly detection [4], image classification [5], [6], and so on. HSI CD is one of the most attractive research applications, which uses dual-temporal HSIs obtained in the same region at different times to determine whether the land cover has changed [7], and it is widely applied in land cover transitions [8], environment monitoring [9], and disaster assessment [10]. Traditional HSI-CD methods mainly include algebra-based methods, transformation-based methods, and classification-based methods. The image algebra methods evaluate the intensity of the change of ground objects by performing mathematical calculations on the relevant band directly [11]. Change vector analysis (CVA) [12] is a typical algebra-based method, which identifies changes by calculating the difference in the spectral vectors of the corresponding pixels of dual-temporal images. However, it is difficult to select an appropriate threshold that directly affects the accuracy of detection [13]. Transformation-based methods convert the original image to another space or domain by mathematical transformation to extract features that are more conducive to CD [14], such as principal component analysis (PCA) [15], multivariate alteration detection (MAD) [16], and iteratively reweighted MAD (IR-MAD) [17]. Such methods show excellent performance in reducing spectral redundancy, but lead to the loss of spectral information. Classification-based methods first classify dual-temporal HSIs, respectively, and analyze the CD result by comparing the classification results [18], such as support vector machine (SVM) [19] and k-nearest neighbors (KNN) [20]. Although such methods transform the CD problem into a classification problem, they ignore the temporal correlation between multitemporal images. Traditional CD methods fail to utilize the spectral and spatial information on account of only extracting shallow

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texture features that cannot represent targets well, which influences the accuracy of HSI CD to some extent.

In recent years, deep learning (DL) has been widely used in HSI CD [21]. Compared with traditional machine learning CD methods, DL methods have powerful representational learning ability and excavate deep features better [22]. The convolutional neural network (CNN) [23], as one of the classic network architectures, has gained great popularity in the CD field and achieved remarkable results [24]. Wang et al. [25] designed a general end-to-end 2-D CNN framework, named GETNET. Methods based on convolutional networks have continued to evolve in the field of CD, with the core challenge lying in efficiently extracting joint features while maintaining precise spatial localization [26], [27]. To address this challenge, research has primarily advanced in two directions. First, constructing complex temporal-spatial-spectral joint feature extraction architectures, such as combining the hidden states of different paths in ConvLSTM to collaboratively model multidimensional dependencies [28]. Second, designing lightweight convolutional structures and operators aims to enhance model efficiency while preserving fine-grained information [29]. Such research employs innovative approaches like artificially padded convolutions or composite convolutional operators to minimize spatial positional information loss caused by downsampling during feature extraction. This enables more precise localization of change regions and better retention of fine details [30]. In addition to CNN-based methods, graph convolution networks (GCNs) [31], [32], recurrent neural networks (RNNs) [33], [34], and other DL methods are also widely used in the field of HSI CD. GCNs can effectively extract spectral–spatial features of irregular regions on account of aggregating adjacent nodes in the graph. Qu et al. [35] proposed a network architecture based on GCNs, named D^2AGCN. This method utilizes the superpixel sparse graph store technique and performs well in detecting subtle changes. RNN is well known to be good at processing sequential data [36]. A novel recurrent CNN (ReCNN) [37] was designed to model the temporal correlation between dual-temporal images, unlike most of the algorithms that use fairly simple operations like image differencing or stacking. However, these methods often require the design of more complex architectures, which increases the number of parameters and makes model training difficult. Wang et al. [38] introduced an end-to-end residual self-calibrated network (RSCNet), which is able to reduce complexity while fully utilizing spatial information.

Currently, Transformer-based methods have achieved dominance in relevant fields due to their powerful attention mechanisms [39]. The core advantage of these approaches lies in their ability to jointly model complex spatial, spectral, and temporal dependencies within hyperspectral data [2]. In particular, Wang et al. [40] introduced a joint spectral, spatial, and temporal Transformer, named SST-Former, which uses Transformer encoders (TEs) of spectral bands and spatial pixels to extract the sequence information. This method helps to extract useful CD features between the current HSI pairs and obtain the binary CD result. To further enhance feature discriminative power, another class of methods focuses on deep excavation and enhancement of spectral features [41]. This is achieved through synergistic operations like differential analysis and dot products to comprehensively dissect spectral dependencies, or by leveraging internal spectral similarity modeling to strengthen identification of genuinely changing regions [42], [43]. In addition, some studies attempt to optimize feature representations through architecture search and adaptive attention mechanisms [44].

Transformer-based CD methods demonstrate significant advantages in effectively representing multilevel features and enhancing model discrimination capabilities [45]. To fully leverage this potential, existing research has designed various feature enhancement strategies across different dimensions [46]. One category of studies focuses on the deep integration of semantic and prior information [47]. Another category of approaches concentrates on strengthening the interaction between multitemporal and multiscale features [48], [49]. By designing pyramid-like cross-temporal attention mechanisms or cross-encoders, these methods facilitate extensive information exchange among feature maps from different time periods and scales, thereby enhancing the ability to capture diverse change patterns [50]. Furthermore, to improve model robustness across diverse scenarios, other approaches concentrate on adaptive feature adjustment and contextual aggregation [51]. By leveraging the TE to adaptively reconstruct feature relationships, suppress noise interference, and aggregate compact spatio-temporal contextual information, these methods effectively mitigate false change phenomena and enhance model generalization capabilities [52], [53].

The Transformer architecture has garnered significant attention for its exceptional ability to model global dependencies, effectively mitigating the loss of long-range information common in traditional models [54]. To fully exploit this capability, numerous studies have focused on designing mechanisms that synergistically capture both local details and global context [55], [56]. For instance, some approaches enhance feature representations by introducing strategies that combine global and local attention mechanisms. Li et al. [57] designed a temporal memory module that detects the semantic information from high to low levels of multitemporal image pairs. Although Transformers can efficiently capture dependencies at arbitrary distances through their unique attention mechanism, they neglect edge features in changing regions, leading to degraded quality in change maps [58], [59]. For detecting edge changes with pixel-level precision, position encoding struggles to accurately represent sub-pixel positional variations, resulting in the model's weak sub-pixel localization capability for edges [60].

However, although these Transformer-based methods have achieved outstanding detection performance, there are still some problems to be solved [61]. The vast majority of existing DL methods place equal importance on all spectral information when extracting spectral features, making it difficult to effectively extract key features. Moreover, these methods primarily focus on spatial-domain difference features while neglecting the potential information that may exist in the frequency domain [62]. In addition, uncertainty theory has found widespread application in the field of image processing, typically categorized into two types: aleatoric uncertainty and

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epistemic uncertainty. Within the field of CD, this uncertainty is mainly manifested in the difficulty of accurately identifying object boundaries [63].

To address these challenges, this article proposed a spectral frequency-induced edge enhancement Transformer (SFIEET) network based on the spectral frequency and spatial-domain fusion (SFSDF) module, the sharpening and Laplace-guided edge enhancement (SLGEE) module, and depthwise convolution-embedded lightweight Transformer (DW-ELT). The SFSDF module mainly divides the HSI into several low-frequency and high-frequency groups according to the spectral curve by a novel spectral frequency grouping based on self-attention (SFGSA) mechanism using the discrete cosine transform (DCT). The low-frequency group pays more attention to the contour and background information of the image, while the high-frequency group pays more attention to the detailed structure of the image, and effectively layers the image by combining the low-frequency and high-frequency, so as to obtain the key change information. SLGEE implements a specialized image sharpening operation by combining maximum pooling and mean pooling to generate locally distinctive self-attentive feature maps that enhance image edges. Subsequently, the enhanced features are fed into the DW-ELT module to further strengthen feature representation. DW-ELT adopts a lightweight architecture that reduces model parameters through depthwise convolution (DWConv) and improves generalization by introducing a bias term. Finally, the refined features are processed by a classifier to produce the CD results.

The main contributions of this article are summarized as follows.

1) The SFSDF module is designed to dynamically focus on channels containing weak semantic information by introducing an adaptive frequency-domain attention mechanism based on group DCT. This mechanism effectively extracts frequency-domain information with latent variation characteristics. At the same time, the module integrates multiscale contextual spatial information to capture morphological changes such as texture fragmentation, significantly enhancing its ability to recognize structural variations.
2) The SLGEE module combines multiscale pooling operations with the Laplacian operator to achieve cross-spectral band edge-feature fusion. Through multiscale pooling, it produces localized and distinctive self-attention feature maps that effectively enhance local boundary details. In addition, a Laplacian filter is independently applied to each channel to further improve the detailed representation of boundaries. This design not only enables precise boundary localization but also provides more reliable detection in ambiguous regions, improving boundary clarity and positional accuracy in areas of change.
3) The DW-ELT module incorporates depthwise separable convolutions into the Transformer architecture to strengthen local feature modeling while reducing computational complexity through parameter-efficient optimization. Gated linear units (GLUs) further enhance nonlinear representations, making features in changing regions more discriminative and thereby improving overall CD performance. Extensive experiments conducted on three public datasets demonstrate the superior effectiveness and robustness of the proposed method.

The remainder of this article is organized as follows. In Section II, a detailed description of the SFIEET network is provided. Section III presents extensive experimental results and analytical discussions. Section IV concludes with a summary for future work.

II. Proposed Method

This section provides an overview of the proposed SFIEET architecture and its constituent modules, as illustrated in Fig. 1. First, the dual-temporal HSIs are divided into small 3-D patches, which serve as inputs to the SFSDF module. Subsequently, spectral–spatial features are jointly extracted through frequency-domain analysis and multiscale spatial convolutions. Next, 3-D pooling and Laplacian filtering are employed to enhance local edge features. Finally, the enhanced features are passed to the DW-ELT module for comprehensive feature learning, and the classifier head performs the final CD.

A. Spectral Frequency and Spatial-Domain Fusion

Initially, bitemporal HSIs $T_1, T_2 \in \mathbb{R}^{c\times h\times w}$ are provided, where c represents the number of spectral channels, and h and w, respectively, represent the height and width of the HSIs. Then, the images T_1, T_2 are divided into patches with the size of $S \times S$, denoted as $P \in \mathbb{R}^{c\times S\times S}$ where patches from T_1 are denoted as P_1 and patches from T_2 are denoted as P_2. Subsequently, these bitemporal patches will be simultaneously fed into the SFSDF module to learn mixed spectral and spatial features. The SFSDF module is shown in Fig. 2. There are four input branches arranged from top to bottom; the first three are designed to extract spatial features at different scales, while the fourth focuses on extracting spectral features in the frequency domain. Spatial feature information at different scales is extracted by using multiscale convolutions (MConv) with different kernel sizes, using a 5×5 convolution ($\text{kernel}_1 = 5, \text{stride}_1 = 1, \text{padding}_1 = 2$), a 3×3 convolution ($\text{kernel}_2 = 3, \text{stride}_2 = 1, \text{padding}_2 = 1$), and a 1×1 convolution ($\text{kernel}_3 = 1, \text{stride}_3 = 1, \text{padding}_3 = 0$). Then, three spatial features at different scales are superimposed to obtain a new multiscale spatial feature $F_{\text{Spa}} \in \mathbb{R}^{C\times H\times W}$, expressed as

$$\begin{aligned} F_{\text{Spa}} = &\text{BN}\left(\text{Conv}\left(P, \text{kernel}_1, \text{stride}_1, \text{padding}_1\right)\right) \\ &+ \text{BN}\left(\text{Conv}\left(P, \text{kernel}_2, \text{stride}_2, \text{padding}_2\right)\right) \\ &+ \text{BN}\left(\text{Conv}\left(P, \text{kernel}_3, \text{stride}_3, \text{padding}_3\right)\right). \end{aligned} \tag{1}$$

To extract high- and low-frequency spectral features, the patch is first shallowly extracted by convolution, noting the shallow feature as Group $\in \mathbb{R}^{C\times H\times W}$. Subsequently, the feature Group is divided into a number of equal-sized parts by channel dimension. Denote $[\text{Group}^1, \text{Group}^2, \ldots, \text{Group}^n]$ as the groups, where $\text{Group}^i \in \mathbb{R}^{C'\times H\times W}$, $i \in \{1, 2, \ldots, n\}$, $C' = C/N$. C' represents the number of channels in each group, and N stands for the number of divided groups. For

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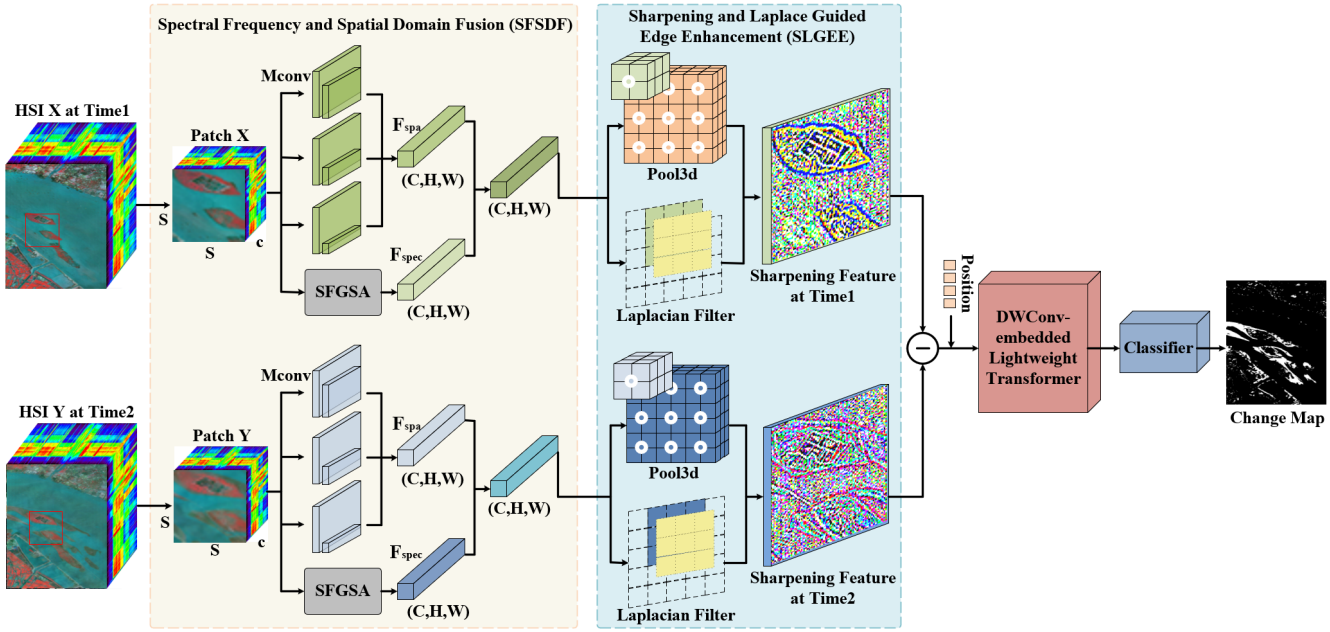


Fig. 1. Overall architecture of the proposed SFIEET for HSI CD.

each group, a set of DCT frequency components is assigned. Thus, we have

$$\text{Freq}_i = \text{DCT}^{u^i,v^i}\left(\text{Group}^i\right) = \sum_{x=1}^{H}\sum_{y=1}^{W} \text{Group}^i_{:,x,y} B^{u^i,v^i}_{x,y} \quad \text{s.t.} \quad i \in \{1,2,\ldots,n\} \tag{2}$$

where $[u_i, v_i]$ $(1 \le u_i \le H, 1 \le v_i \le W)$ are the frequency component 2-D indices corresponding to Group. Lower index values correspond to lower frequencies that contain the primary information of the image. In contrast, higher frequencies correspond to finer details within the image.

$\text{Freq}_i \in \mathbb{R}^{C'}$ is the C'-dimensional vector after compression. The whole compression vector can be obtained by concatenation

$$\begin{aligned}\text{Freq} &= \text{compress}\left(\text{Group}\right)\\ &= \text{Concat}\left(\left[\text{Freq}_1, \text{Freq}_2, \ldots, \text{Freq}_n\right]\right)\end{aligned} \tag{3}$$

where $\text{Freq} \in \mathbb{R}^C$ is the obtained multispectral vector.

Freq passes through a fully connected (FC) layer and sigmoid activation function to obtain the spectral frequency self-attention Att. FC consists of two linear layers and a ReLU activation function. Finally, the input feature Y and the self-attention Att are multiplied by element points ($\odot$) to obtain the high- and low-frequency spectral features $F_{\text{Spec}} \in \mathbb{R}^{C\times H\times W}$. The whole spectral frequency self-attention framework can be written as

$$\begin{aligned}F_{\text{Spec}} &= \text{Att} \odot Y\\ &= \text{sigmoid}\left(\text{FC}\left(\text{Freq}\right)\right) \odot Y\\ &= \text{sigmoid}\left(\text{Linear}\left(\text{ReLU}\left(\text{Linear}\left(\text{Freq}\right)\right)\right)\right) \odot Y.\end{aligned} \tag{4}$$

In order to obtain the spectral–spatial feature, the spectral feature F_{Spec} is first connected with the spatial feature F_{Spa} in the channel dimension to obtain the feature $F_{\text{cat}} \in \mathbb{R}^{2C\times H\times W}$. Next, since the spectral and spatial information of the connected features is dispersed, a convolution operation is applied to fully integrate these features while simultaneously reducing the number of channels to C, thus obtaining the mixed feature $F_{\text{mix}} \in \mathbb{R}^{C\times H\times W}$. In order to minimize the impact of network degradation on the results, an additional improved residual term is introduced to obtain the final mixed spectral–spatial feature output $F_{\text{out}} \in \mathbb{R}^{C\times H\times W}$. The process of feature fusion can be defined as

$$\begin{aligned}F_{\text{cat}} &= \text{Concat}\left(F_{\text{Spec}}, F_{\text{Spa}}\right)\\ F_{\text{mix}} &= \text{BN}\left(\text{Conv}\left(\text{ReLU}\left(\text{BN}\left(\text{Conv}\left(F_{\text{cat}}\right)\right)\right)\right)\right)\\ F_{\text{out}} &= \text{BN}\left(\text{Conv}\left(F_{\text{mix}} + \text{Conv}\left(F_{\text{cat}}\right)\right)\right).\end{aligned} \tag{5}$$

B. Sharpening and Laplace-Guided Edge Enhancement

The edge portion of the detection is often interfered with by neighboring pixels, which leads to misdetection and omission. For this reason, we designed the SLGEE module to mitigate this unfavorable interference. As illustrated in Fig. 3, SLGEE consists of two branches, a sharpening branch and a Laplace enhancement branch. In the sharpening branch, F_{out} is processed by both mean pooling and maximum pooling. The resulting pooled features are concatenated along the channel dimension and then combined through elementwise summation to obtain $z_1 \in \mathbb{R}^{2\times H\times W}$ and $z_2 \in \mathbb{R}^{1\times H\times W}$, respectively. Max pooling emphasizes boundary regions, while mean pooling, although it smooths the image and slightly blurs edges, effectively suppresses noise interference and prevents noise from disrupting boundary detection. By integrating the two, a local distinctive attention map $z_{\text{local}} \in \mathbb{R}^{1\times H\times W}$ is generated, which helps to accurately locate boundary regions. Guided by this attention, the model effectively highlights local distinctive boundaries, yielding the refined feature

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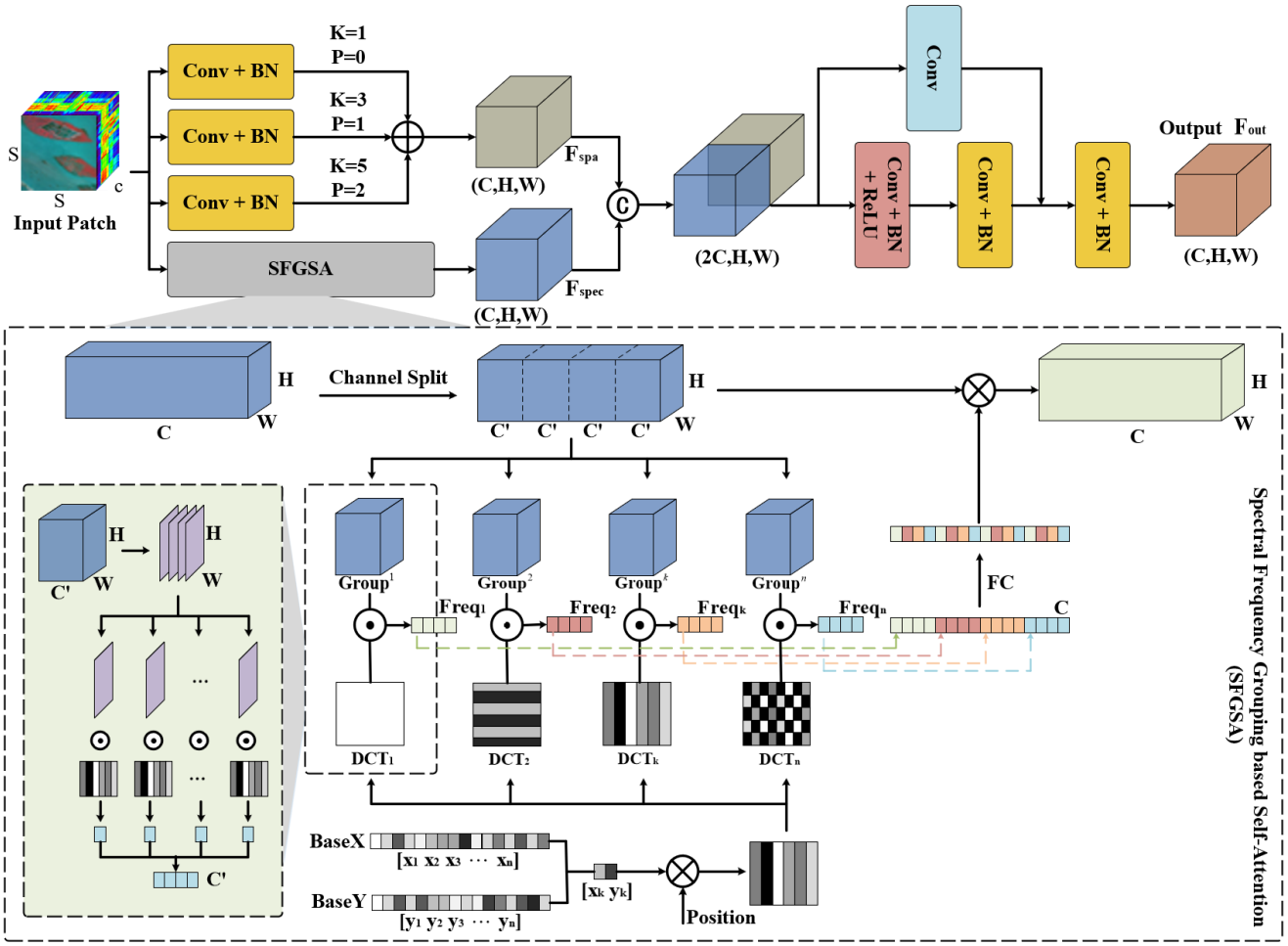


Fig. 2. Architecture of the SFSDF.

representation $F_{\text{local}} \in \mathbb{R}^{C\times H\times W}$. The local distinctive attention calculation process can be summarized as follows:

$$\begin{aligned} z_1 &= \text{Concat}\left(\text{AvgPool3d}\left(F_{\text{out}}\right), \text{MaxPool3d}\left(F_{\text{out}}\right)\right) \\ z_2 &= \text{AvgPool3d}\left(F_{\text{out}}\right) + \text{MaxPool3d}\left(F_{\text{out}}\right) \\ z_{\text{local}} &= \text{BN}\left(\text{Conv}\left(\text{Conv}\left(z_1\right) + z_2\right)\right) \\ F_{\text{local}} &= F_{\text{out}} \odot z_{\text{local}}. \end{aligned} \tag{6}$$

In the Laplace enhancement branch, the Laplacian operator $\mathcal{L}$ is chosen to sharpen the feature representation. The Laplacian filter detects the edges of an image by performing a second-order derivative on the luminance, thereby emphasizing regions with abrupt gray-level transitions and enhancing fine edge details. Since the image is stored in a discrete form, the differentiation process is implemented using convolution. We partition the input features along the channel dimension into 2-D features while preserving spatial resolution. This operation is applied independently to each 2-D feature within the hybrid feature, after which the generated edge-enhanced features are stitched together along the channel dimension to form $F_{\text{Laplace}} \in \mathbb{R}^{C\times H\times W}$. This can be expressed as

$$\begin{aligned} F_{\text{Laplace}} &= \mathcal{L}\left(F_{\text{out}}\right) \\ &= \mathcal{L}\left[F_{\text{out}}^1, F_{\text{out}}^2, \ldots, F_{\text{out}}^n\right]. \end{aligned} \tag{7}$$

Since each channel is processed independently, the obtained edge features are further convolved and normalized to restore the interchannel correlations, yielding the feature $F'_{\text{Laplace}} \in \mathbb{R}^{C\times H\times W}$

$$F'_{\text{Laplace}} = \text{BN}\left(\text{Conv}\left(\text{BN}\left(\text{ReLU}\left(\text{Conv}\left(F_{\text{Laplace}}\right)\right)\right)\right)\right). \tag{8}$$

After denoising and edge enhancement are performed separately, the two resulting features, F_{local} and F'_{Laplace}, are summed to obtain the enhanced hybrid feature $F_{\text{EN}} \in \mathbb{R}^{C\times H\times W}$

$$F_{\text{EN}} = F_{\text{local}} + F'_{\text{Laplace}}. \tag{9}$$

C. DWConv-Embedded Lightweight Transformer

The structure of the DW-ELT module is illustrated in Fig. 4. In conventional Transformer architectures, positional coding is typically employed to enable token-based processing. However, this encoding partitions the image into discrete patches, thereby disrupting its inherent local spatial continuity. To alleviate this structural degradation, we incorporate DWConv to extract local information, which is defined as

$$\text{LDW}\left(X\right) = \text{DWConv}\left(X\right) + X \tag{10}$$

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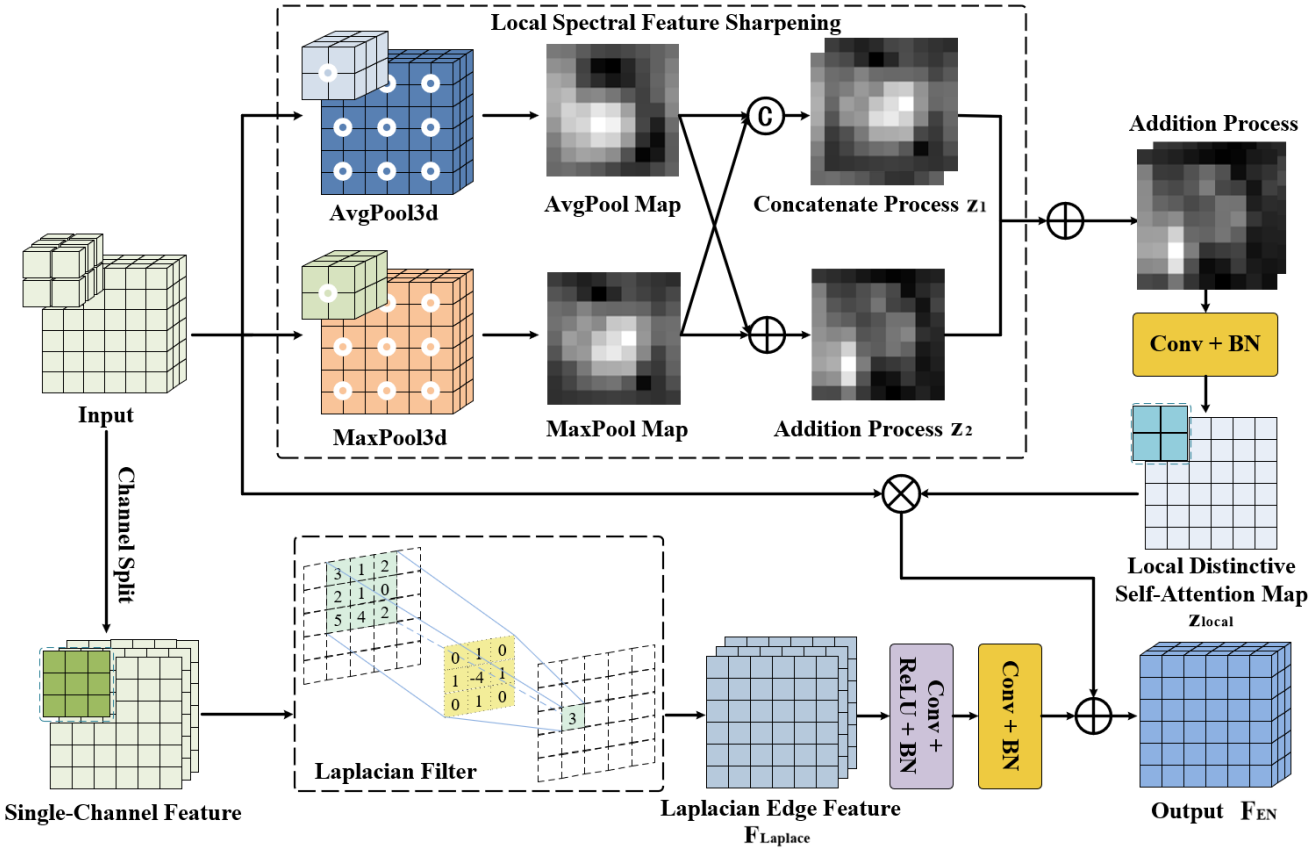


Fig. 3. Architecture of the SLGEE.

where $X \in \mathbb{R}^{C\times H\times W}$, $H \times W$ is the spatial resolution of the input at the current stage, and C indicates the feature dimension.

To facilitate attention computations, tensor X is reshaped from $C \times H \times W$ to $n \times d$. In the original self-attention module, the input $X \in \mathbb{R}^{n\times d}$ is linearly transformed the query $Q \in \mathbb{R}^{n\times d_k}$, the key $K \in \mathbb{R}^{n\times d_k}$, and the value $V \in \mathbb{R}^{n\times d_v}$, where $n = H \times W$ is the number of patches and d is equal to the feature dimension C. The notations d, d_k, and d_v are the dimensions of input, key (query), and value, respectively,

$$\text{Attention}\,(Q,K,V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \tag{11}$$

To reduce the computational complexity, we use a $k \times k$ DWConv with stride k to reduce the size of the key K and the value V as a way to speed up the process of attention computation

$$\begin{aligned} K' &= \text{DWConv}\,(K,k) \\ V' &= \text{DWConv}\,(V,k) \end{aligned} \tag{12}$$

where $K' \in \mathbb{R}^{n/k^2\times d_k}$ and $V' \in \mathbb{R}^{n/k^2\times d_v}$ are obtained by DWConv of K and V, respectively. In addition, a relative position bias $B \in \mathbb{R}^{n\times n/k^2}$ is added to each self-attention module, and the lightweight attention is defined as

$$\text{LightAttention}\left(Q,K',V'\right) = \text{Softmax}\left(\frac{QK'^T}{\sqrt{d_k}} + B\right)V'. \tag{13}$$

The input is passed through a dual-branch multilayer perceptron (DBMLP), with one branch using the ReLU activation function and the other without additional changes. Subsequently, the results of the two branches are multiplied and passed through the linear layers again to get the final output

$$\text{DBMLP}\,(X) = \text{Linear}\,(\text{ReLU}\,(\text{Linear}\,(X)) * \text{Linear}\,(X)). \tag{14}$$

D. Classifier Head

To achieve the final change-region detection, a classifier header is designed for distinguishing binary changes. After processing through the DW-ELT module, a long sequence of features containing change information is obtained. This sequence is first reshaped into a 3-D feature representation; after which the feature dimension is expanded via a convolution operation, with the output dimension determined by the parameter dim in the classifier head. To mitigate the impact of outlier values on model performance, mean pooling is applied to the expanded features, thereby enhancing their robustness. Subsequently, the pooled features are flattened into a 1-D vector of length dim and mapped to the change category space through an FC layer, ultimately achieving binary CD.

The process of the proposed SFIEET network is shown in Algorithm 1.

III. Experiments and Analysis

In this section, the effectiveness of the proposed method is evaluated through comprehensive experiments conducted on

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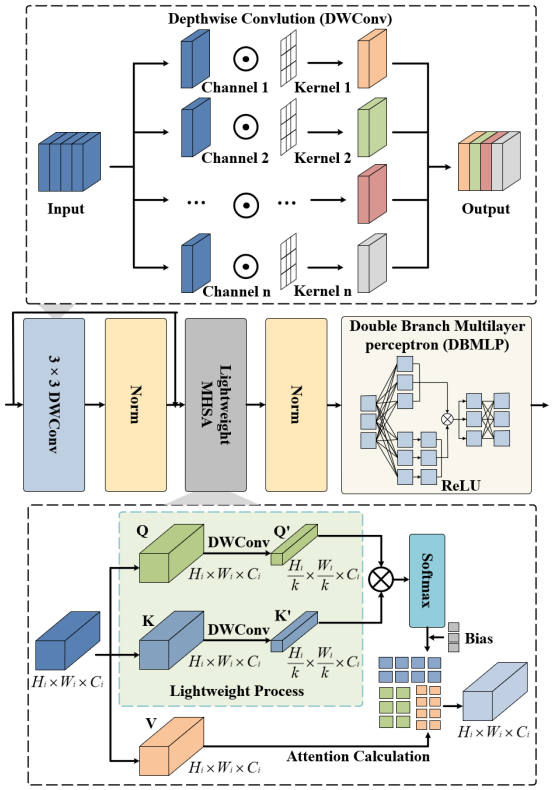
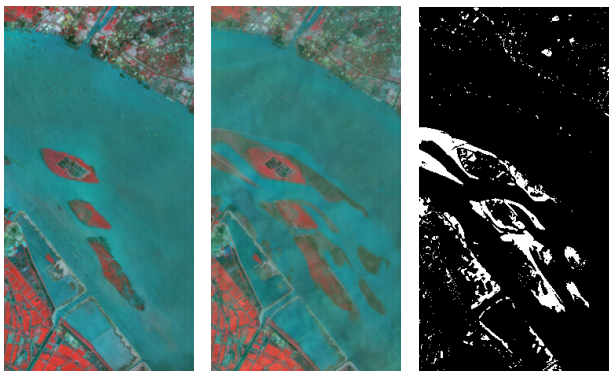


Fig. 4. Architecture of the DW-ELT.



(a) (b) (c)

Fig. 5. River dataset. (a) River on May 3, 2013. (b) River on December 31, 2013. (c) Ground truth.

three publicly available datasets. First, the three datasets used in the experiments are described in detail. Next, comparative experiments are carried out under different configurations and parameter settings, and the influence of each parameter on the model's performance is analyzed based on the obtained results. The proposed method is then compared with several traditional approaches as well as recent state-of-the-art DL methods. Finally, a series of ablation experiments is performed to verify the contribution and effectiveness of each module within the proposed framework.

A. Dataset Description

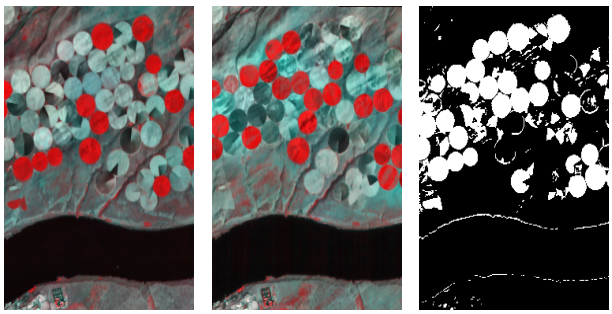
We evaluate the proposed method on three widely-used dual-temporal HSI datasets, each capturing distinct types of land cover changes.

Algorithm 1 SFIEET Network for HSI CD

Input: Bitemporal HSIs $T_1, T_2 \in \mathbb{R}^{C*h*w}$ and label $Y \in \mathbb{R}^{h*w}$; patch size S; training sample rate $\alpha\%$.

Output: Predicted labels of the test set.

Set batch size bs = 32; learning rate lr = 5e-4; epochs E = 120.
for i = 1 to E **do**
  Perform multiscale convolutions on patches to acquire spatial features F_{Spa} by Eq. (1).
  Perform SFGSA on patches to acquire spectral features F_{Spec} by Eq. (2)-(4).
  Fuse F_{Spa} and F_{Spec} to get spectral–spatial features F_{out} via Eq. (5).
  Apply mean pooling and max pooling on F_{out} to obtain z_1 and z_2.
  Derive local attention map z_{local} from z_1 and z_2, then compute sharpening feature F_{local} through Eq. (6).
  Perform channel-wise Laplacian filtering on F_{out} to get the feature $F'_{Laplace}$ by Eq. (7)-(8).
  Obtain the enhanced hybrid features F_{EN} by superimposing F_{local} and $F'_{Laplace}$ by Eq. (9).
  Feed the feature F_{EN} into the DW-ELT for feature learning by Eq. (10)-(14).
  Input the DW-ELT token to the classifier head to obtain the category labels.
end for
Employ the trained model to perform predictions on the test set.



(a) (b) (c)

Fig. 6. USA dataset. (a) USA on May 1, 2004. (b) USA on May 8, 2007. (c) Ground truth.

1) The dual-temporal HSIs used for CD in the river dataset were collected on May 3 and December 31, 2013, from a river area in Jiangsu Province, China. The sensor used for shooting is Earth Observing-1 (EO-1) Hyperion, with a spectral range of 0.4–2.5 μm, a spectral resolution of 10 nm, and a spatial resolution of 30 m. The size of the image in the dataset is 463×241 pixels, the source data have 242 spectral bands, and after removing some noise bands, 198 spectral bands are obtained for detection tasks. The main change in the river dataset is reflected in the change of the river course. The pseudo-color composite image and ground-truth map are illustrated in Fig. 5.

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2) The dual-temporal HSIs used for CD in the USA dataset were collected on May 1, 2004 and May 8, 2007 from the agriculture area in Hermiston City, Umatilla County, Oregon, USA. The sensor used is EO-1 Hyperion. The size of the image in the dataset is 307 × 241 pixels, and after removing some noise bands, 156 bands are obtained for detection tasks. The changes in this dataset are mainly the large cultivated land changes and relatively fine riverbank line changes. The pseudo-color composite image and ground-truth map are illustrated in Fig. 6.
3) The dual-temporal HSIs used for CD in the Farmland dataset were collected on May 3, 2006 and April 23, 2007, from farmland near Yancheng City, Jiangsu Province, China. The sensor used is EO-1 Hyperion. The size of the image in the dataset is 450 × 140 pixels, the source data have 242 bands, and after removing some noise bands, 155 bands are obtained for detection tasks. The changes in this dataset mainly reflect the cultivation area and crop changes, which are relatively concentrated. The pseudo-color composite image and ground-truth map are illustrated in Fig. 7.

B. Experimental Setup

1) Configurations: To ensure the fairness of the experiments, both the proposed method and all comparison approaches were implemented using the PyTorch framework. Training and testing were conducted on a server equipped with an Intel Xeon Silver 4314 2.40-GHz CPU, 192-GB RAM, and two NVIDIA GeForce RTX 4090 GPUs with 24 GB of memory each.

2) Evaluation Indicators: We quantitatively evaluated the CD performance of all methods using five well-established evaluation metrics. Of these indicators, overall accuracy (OA) is the number of correctly categorized samples as a proportion of the number of all samples. The kappa coefficient is a metric used for consistency testing to discern whether the predicted results are consistent with the actual results. Precision reflects the reliability of the model's predictions, especially in the determination of positive class samples. Recall is one of the core metrics reflecting the performance of the model, which directly reflects the model's ability to capture positive class samples and is suitable for scenarios where leakage detection is costly. $F1$-score ($F1$) is a core metric for comprehensively evaluating the performance of models, which balances precision and recall through the harmonic mean.

C. Parameter Analysis

1) Patch Size: In order to investigate the impact of patch size on the performance of CD, all other parameters were kept constant while selecting candidate patch sizes of 7, 9, 11, 13, and 15. It is obvious that the three datasets achieved optimal performance when the patch size was set to 9, as shown in Fig. 8(a).

2) Batch Size: Batch size affects the training speed, accuracy, and generalization ability of the network. A larger batch size requires traversing more data in the training set in each iteration, which can lead to excessively long training times. Conversely, a batch size that is too small may cause unstable convergence due to insufficient data representation, leading to high noise and degraded training performance. To evaluate the effect of batch size, candidate values of 16, 32, 64, 128, and 256 were tested while keeping all other parameters fixed. As shown in Fig. 8(b), our method achieved optimal performance when the batch size was set to 32.

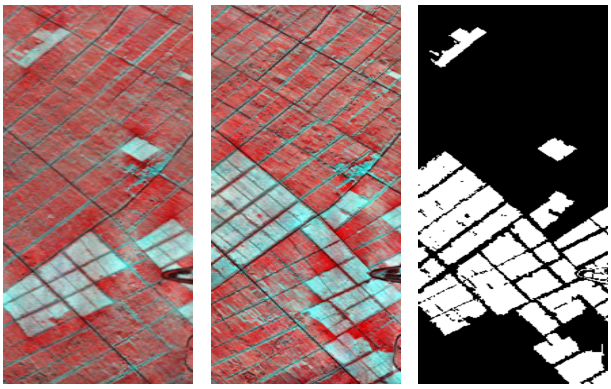


Fig. 7. Farmland dataset. (a) Farmland on May 3, 2006. (b) Farmland on April 23, 2007. (c) Ground truth.

3) Learning Rate: Learning rate has a significant influence on the training speed and convergence of the model. A learning rate that is too small may cause slow convergence, resulting in excessively long training times. Conversely, an overly large learning rate can accelerate training to some extent but often prevents the model from reaching an optimal local solution. Determining an appropriate learning rate, therefore, requires multiple experiments to balance convergence speed and OA. In our experiments, candidate learning rates of e-5, 5e-5, e-4, 5e-4, and e-3 were tested. As shown in Fig. 8(c), the best OA was achieved when the learning rate was set to 5e-4.

4) Percentages of Training Samples: To evaluate the robustness of the proposed method, we conducted tests on the CD accuracy of all methods under different sampling rates across the three datasets. We randomly selected 2%, 5%, 15%, 25%, and 35% of samples for training. Fig. 9 shows the OA of all methods at different sampling rates. It is observed that, with a very limited number of training samples, some methods experience a significant decrease in detection accuracy, while the proposed method maintains a relatively high accuracy and exhibits strong robustness.

5) Group of DCT: For the SFGSA module, the selection of the number of DCT groups has a substantial impact on the CD performance. As illustrated in Fig. 10, using too few or too many groups adversely affects the OA. To evaluate this effect, the number of groups was varied among the candidate values of 2, 4, 8, 16, and 32, while keeping all other parameters fixed. When the number of groups was set to 16, our method achieved optimal performance across all three datasets.

D. CD Results and Analysis

To verify the superiority of the proposed SFIEET, we compared it with nine state-of-the-art methods, including

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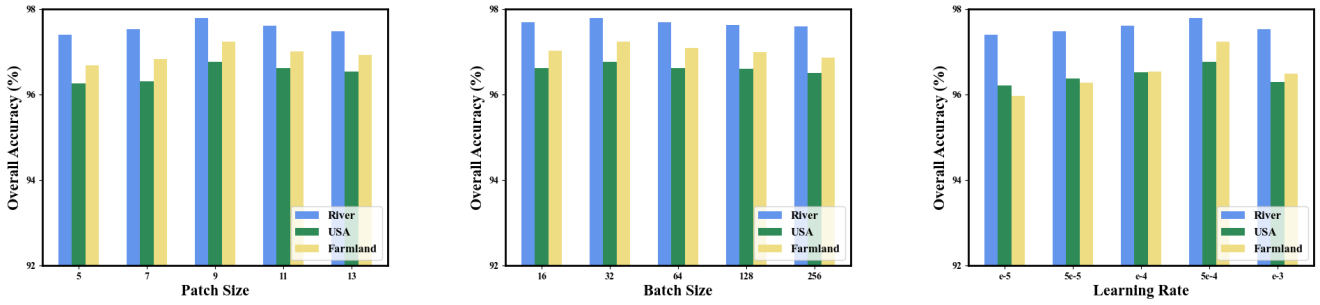


Fig. 8. Impact of different parameters on multiple datasets. (a) Patch size. (b) Batch size. (c) Learning rate.

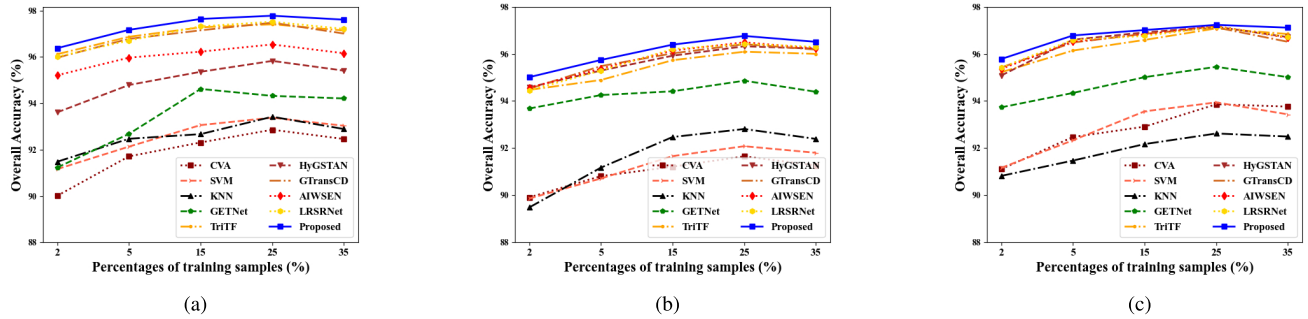


Fig. 9. OA of different models with different training sample percentages. (a) River dataset. (b) USA dataset. (c) Farmland dataset.

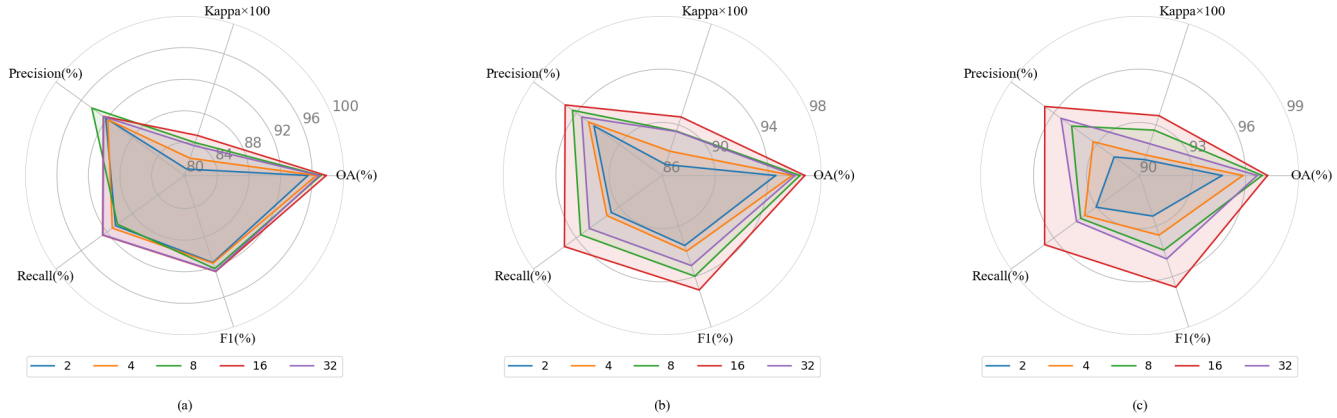


Fig. 10. Impact of different groups of DCT on OA, kappa, precision, recall, and $F1$. (a) River dataset. (b) USA dataset. (c) Farmland dataset.

CVA [12], SVM [19], KNN [20], GETNet [25], TriTF [64], HyGSTAN [65], GTransCD [66], AIWSEN [67], and LRSR-Net [68]. For a fair comparison, all the compared methods adopt the default parameters provided in their work. OA, kappa, precision, recall, and F1 for the proposed method and all comparative methods on the three datasets are shown in Table I, where the best results are highlighted in bold, and the second-best results are underlined. The results demonstrate that the proposed method outperforms all other approaches across all five evaluation metrics. Regarding the parameter settings for the three datasets, the training epochs for river, USA, and farmland were set to 120, 120, and 150, respectively. The learning rate decay was set to 0.95, with the decay interval adjusted according to the epoch. Throughout the training process, the learning rate decays 20 times. Other parameter values are referenced in the parameter analysis.

Figs. 11–13 illustrate the CD maps of the proposed method and the comparative approaches on all three datasets, where black indicates unchanged areas and white denotes changed areas. The blue boxes highlight locally magnified regions, while the yellow boxes indicate discrepancies between the predicted results and the ground truth—black pixels denote correct predictions, and white pixels represent incorrect ones. It is evident that the CD maps produced by SFIEET exhibit significantly reduced noise and a much closer correspondence to the ground truth. Taking the river dataset as an example, our method achieves markedly superior performance compared with other detection methods, while HyGSTAN and GTransCD also deliver relatively satisfactory detection results.

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TABLE I

QUALITY METRICS OBTAINED BY DIFFERENT HSI-CD METHODS ON THREE DATASETS (BEST RESULTS ARE BOLDED AND 2ND-BEST ARE UNDERLINED)

		CVA	SVM	KNN	GETNet	TriTF	HyGSTAN	GTransCD	AIWSEN	LRSENet	Proposed
River	OA	0.9286	0.9338	0.9342	0.9423	0.9741	0.9583	0.9748	0.9654	0.9752	**0.9778**
	Kappa	0.6593	0.6214	0.7192	0.7414	0.8393	0.8125	0.8382	0.7861	0.8346	**0.8514**
	Precision	0.8405	0.8393	0.8413	0.8405	0.8667	0.9065	0.9142	0.8912	0.9201	**0.9255**
	Recall	0.7372	0.7481	0.7871	0.8376	0.8823	0.9167	0.8298	0.8492	0.9182	**0.9266**
	F1	0.7855	0.7911	0.8133	0.8390	0.8744	0.9116	0.8700	0.8697	0.9191	**0.9260**
USA	OA	0.9165	0.9208	0.9281	0.9486	0.9609	0.9632	0.9641	0.9649	0.9643	**0.9676**
	Kappa	0.8132	0.8342	0.8247	0.8262	0.8743	0.8796	0.8852	0.8968	0.8958	**0.9063**
	Precision	0.9059	0.9151	0.9081	0.9267	0.9461	**0.9577**	0.9542	0.9457	0.9521	0.9503
	Recall	0.8972	0.9042	0.9073	0.9260	0.9465	0.9418	0.9358	0.9468	0.9454	**0.9510**
	F1	0.9015	0.9096	0.9077	0.9263	0.9463	0.9497	0.9449	0.9462	0.9487	**0.9506**
Farmland	OA	0.9386	0.9394	0.9262	0.9545	0.9708	0.9716	0.9712	0.9718	0.9715	**0.9723**
	Kappa	0.8867	0.8526	0.8929	0.8973	0.9338	0.9282	0.9321	0.9335	0.9327	**0.9355**
	Precision	0.9127	0.9174	0.9247	0.9461	0.9647	0.9572	**0.9668**	0.9655	0.9623	0.9663
	Recall	0.9086	0.9162	0.9243	0.9451	0.9638	0.9547	0.9657	0.9648	0.9617	**0.9664**
	F1	0.9106	0.9168	0.9245	0.9456	0.9642	0.9559	0.9662	0.9651	0.9620	**0.9663**

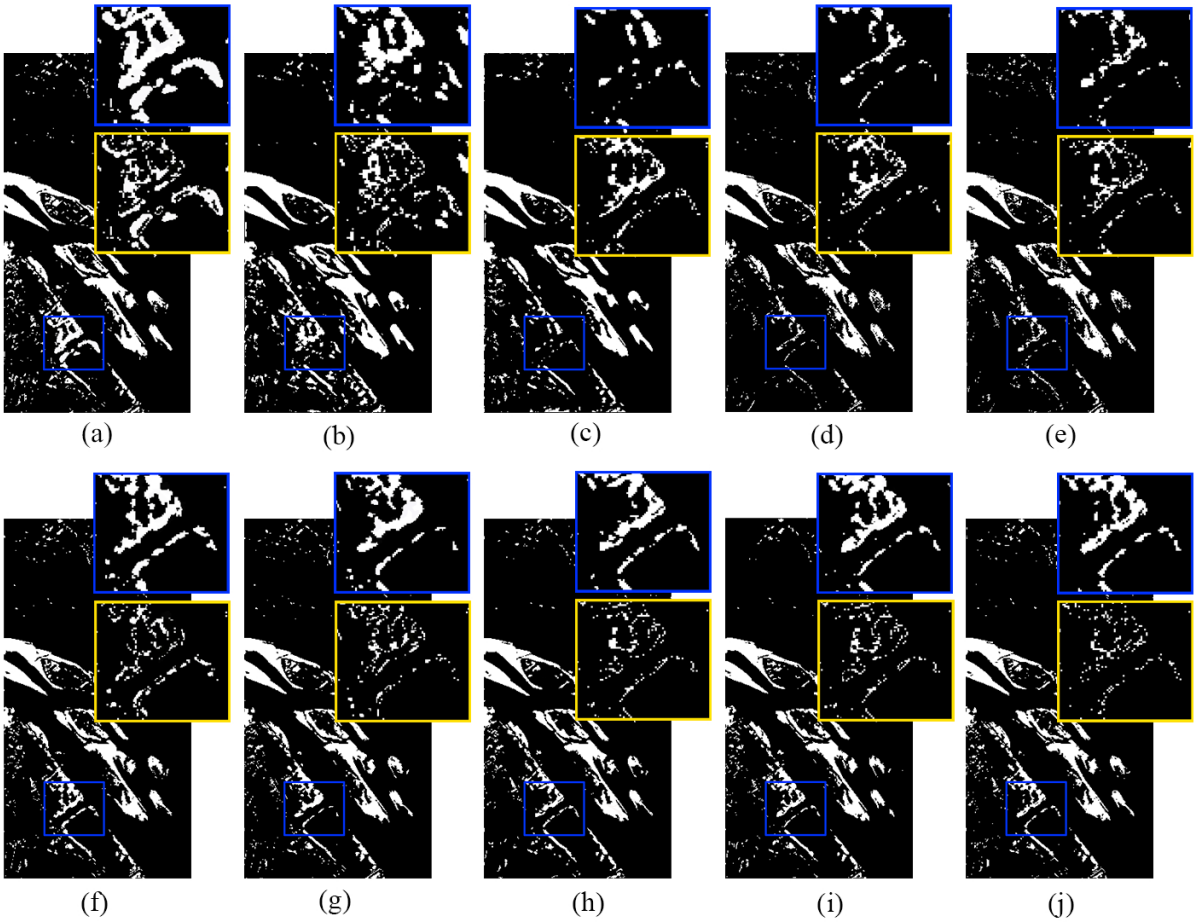


Fig. 11. CD results of different methods on the River dataset. (a) CVA. (b) SVM. (c) KNN. (d) GETNet. (e) TriTF. (f) HyGSTAN. (g) GTransCD. (h) AIWSEN. (i) LRSRNet. (j) Proposed.

1) River Dataset: The prediction results on the river dataset are presented in Fig. 11. The relatively low kappa coefficients observed for traditional CD methods indicate their limited capability to effectively distinguish between changed and unchanged categories. In contrast, DL-based approaches benefit from large-scale sample training, which alleviates this issue to some extent. Nevertheless, achieving highly accurate results on this dataset remains challenging compared with other datasets, likely due to the characteristics of its ground-truth labeling. In particular, the dataset

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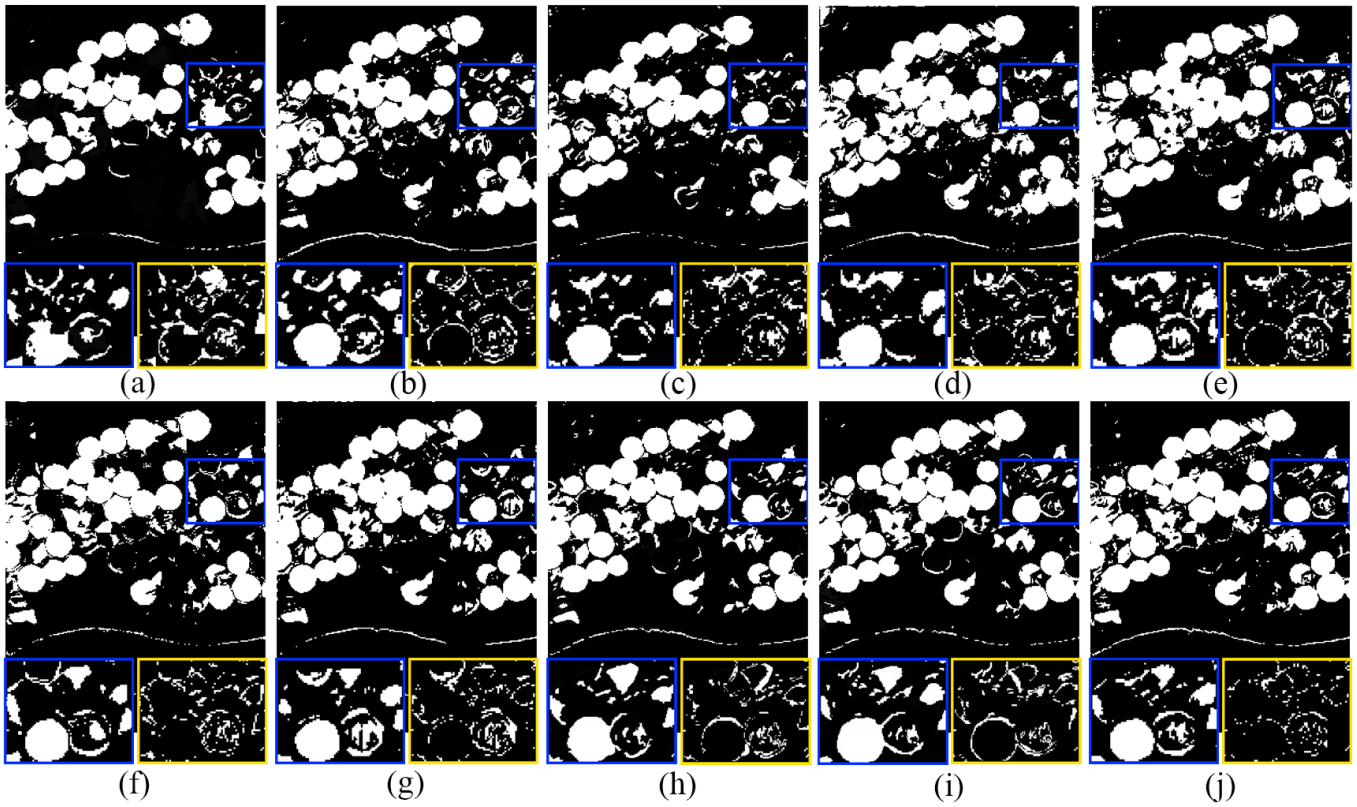


Fig. 12. CD results of different methods on the USA dataset. (a) CVA. (b) SVM. (c) KNN. (d) GETNet. (e) TriTF. (f) HyGSTAN. (g) GTransCD. (h) AIWSEN. (i) LRSRNet. (j) Proposed.

contains a substantial proportion of unchanged pixels, while changed regions occupy only a small fraction. This class imbalance biases the model toward predicting the majority (unchanged) class, thereby constraining improvements in the kappa coefficient. Despite these challenges, our proposed method achieves the best performance on the river dataset across all five evaluation metrics. Compared with the suboptimal results, the OA, kappa, precision, recall, and $F1$-score are improved by 0.25%, 0.0121, 1.13%, 0.99%, and 1.44%, respectively.

2) USA Dataset: As illustrated in Fig. 12, the proposed method captures more subtle variations, exhibits a lower false alarm rate, and produces results that are most consistent with the ground-truth map. In contrast, several DL-based approaches show noticeable clustering artifacts in their predictions, which can be attributed to their limited capability in detecting fine-grained details. In this dataset, the areas of change are relatively concentrated, and our method effectively identifies these regions while accurately delineating their irregular boundaries. Overall, our method achieves the best performance on the USA dataset across four evaluation metrics, i.e., OA, kappa, recall, and $F1$, surpassing the suboptimal results by 0.35%, 0.0211, 0.45%, 0.09%, respectively.

3) Farmland Dataset: Although the three conventional detection methods achieve relatively high OA, they exhibit a considerable number of misdetections, incorrectly classifying the changed and unchanged regions. As shown in Fig. 13, DL-based CD approaches substantially alleviate this issue. It is clear from the maps of the test results that there are very few large areas of misdetection, and only a few at the boundaries. In this dataset, small invariant regions are often embedded within the change regions, and our method occasionally enlarges these invariant regions due to the locally distinctive attention mechanism, which emphasizes changes while de-emphasizing invariants. Consequently, although our method achieves superior overall performance, the improvement relative to other DL methods is modest. In particular, our method achieves the best performance on the Farmland dataset across four evaluation metrics, i.e., OA, kappa, recall, and $F1$, while precision attains suboptimal results.

E. Discussion

1) Ablation Study: To evaluate the contribution of each component in the proposed method, ablation experiments were conducted on the river dataset. The primary components of the proposed method include MConv, SFGSA, SLGEE, and DW-ELT. Evaluation metrics include OA, kappa, precision, recall, and $F1$, and the experimental results are presented in Table II.

In particular, Case 1 shows that the lack of direct extraction of spatial information slightly reduces the OA. However, since surrounding pixels are considered during patch division, MConv has the least impact on the overall results compared to the other three components. Case 2 indicates that the absence of frequency domain spectral features has the most significant effect on performance, reflecting the critical role

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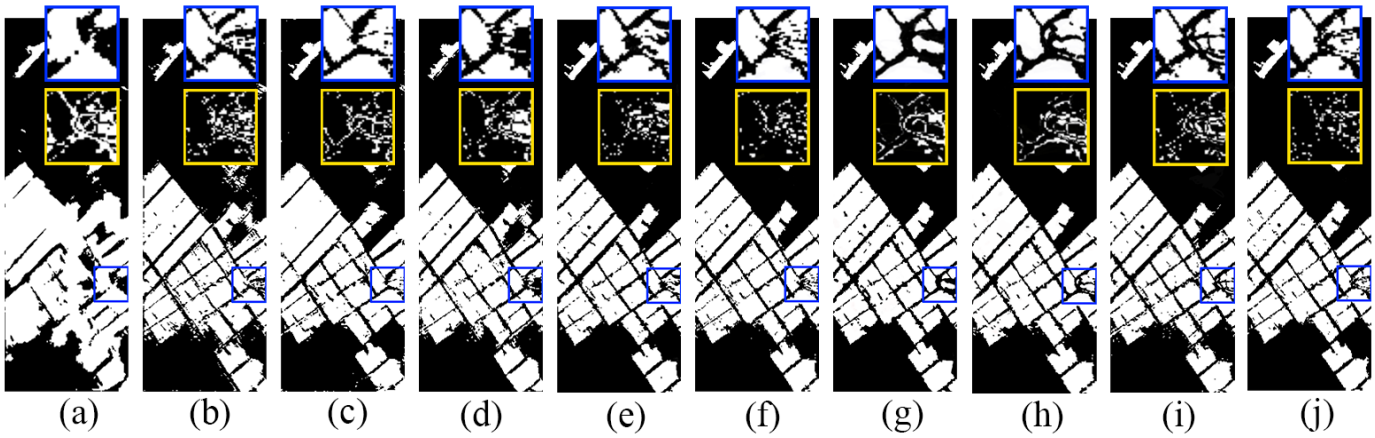


Fig. 13. CD results of different methods on the Farmland dataset. (a) CVA. (b) SVM. (c) KNN. (d) GETNet. (e) TriTF. (f) HyGSTAN. (g) GTransCD. (h) AIWSEN. (i) LRSRNet. (j) Proposed.

TABLE II
ABLATION ANALYSIS OF DIFFERENT COMPONENTS IN THE PROPOSED MODEL ON RIVER DATASET (OPTIMAL RESULTS ARE BOLDED)

Cases	Component				Indicators				
	MConv	SFGSA	SLGEE	DW-ELT	OA	Kappa	Precision	Recall	F1
1	-	√	√	√	0.9761	0.8498	0.9249	0.9260	0.9254
2	√	-	√	√	0.9687	0.8395	0.9184	0.9188	0.9186
3	-	-	√	√	0.9318	0.7589	0.8971	0.8724	0.8864
4	√	√	-	√	0.9724	0.8459	0.9193	0.9202	0.9197
5	√	√	-	-	0.9547	0.8031	0.9014	0.8893	0.8953
6	√	√	√	TE	0.9742	0.8461	0.9216	0.9225	0.9220
7	√	√	√	√	**0.9778**	**0.8514**	**0.9255**	**0.9266**	**0.9260**

of spectral information in HSI CD. Proper frequency domain grouping effectively separates the high- and low-frequency information, enabling the extraction of key features. In Case 3, where edge enhancement was applied directly to the input patch without extracting the spatial spectrum feature, the experiments yielded the poorest results. This demonstrates that feature fusion effectively enhances distinctiveness, thereby enabling the model to accurately recognize change. Case 4 shows that edge enhancement has a nonnegligible impact on detection results. Nevertheless, due to the small proportion of the edge region to the overall image, the impact of SLGEE on the whole is not as obvious as SFGSA from the point of view of each indicator. Case 5 demonstrates that by enhancing edge features and increasing nonlinear representations, the combined use of SLGEE and DW-ELT significantly improves the model's ability to distinguish variations compared to their individual application. In Case 6, DW-ELT was replaced with a standard TE to maintain the overall model structure. The results indicate that the introduction of DWConv and the lightweight design in DW-ELT positively contribute to model performance. In summary, each component of the proposed method plays a constructive role in achieving the final CD performance.

2) Comparison of Parameters and FLOPs: To validate the implementation effectiveness of the DW-ELT module in terms of lightweight design, we calculated the model's parameter count and FLOPs. Concurrently, as a comparative experiment, we replaced the DW-ELT module with a TE module, and substituted the MLP component within the TE with an existing DBMLP. Based on this, we recalculated the model's parameter count and FLOPs. The experimental results demonstrate that the lightweight design reduces the model's parameter count by 23.7% while decreasing FLOPs by 49.71 MB. To further validate the proposed model's efficacy, we also present the parameter counts and FLOPs of the comparison methods. The experimental results are displayed in Table III. Compared to six benchmark methods, our approach achieved the third-best results in terms of both parameter count and FLOPs. It strikes an excellent balance between accuracy and model complexity while maintaining the highest detection accuracy.

TABLE III
COMPARISON OF PARAMS AND FLOPS OF DIFFERENT METHODS

Method	Params(MB)	FLOPs(MB)
GetNet	154.18	90905.67
TriTF	0.19	2421.32
HyGSTAN	0.18	4.27
GTransCD	146.11	7615.81
AIWSEN	12.23	55.98
LRSENet	4.30	359.30
SFIEET(without DW-ELT)	2.48	213.52
SFIEET(with DW-ELT)	1.89	163.81

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IV. Conclusion

In this article, a novel grouped frequency domain Transformer network is proposed for accurate CD in dual-temporal HSIs. A feature extraction and enhancement structure is designed to capture spectral information in the frequency domain and subsequently enhance it through sharpening operations. In particular, spectral features are effectively extracted using grouped DCT and combined with spatial information, preserving spectral sequence characteristics while fully considering spatial contextual correlations. Edge information is further enhanced via sharpening and Laplace filtering, improving the model's ability to recognize fine details. The experimental results on three hyperspectral datasets demonstrate that the proposed method outperforms existing state-of-the-art CD approaches.

Although the proposed SFIEET demonstrates outstanding performance in hyperspectral CD tasks, certain limitations remain. In future research, the grouping frequency domain operation could adaptively select the number of groups for different datasets, thereby effectively addressing diverse types of land cover data. Furthermore, computational efficiency could be enhanced through additional lightweight processing of the modules.

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